Tahmina Sultana Priya

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RESEARCH INTEREST

Machine Learning for Health, Healthcare Data Analytics, Explainable AI, Precision Medicine, Bioinformatics, Large Language Models

EDUCATION

Virginia Polytechnic Institute and State University **Virginia, USA**
Ph.D. in Computer Science *January 2023 – Present*
Supervised by: Dr. Danfeng (Daphne) Yao

Expected Graduation: December 2026
Dissertation: Algorithm-driven Precise and Explainable Prognosis for Complex Metabolic Disorders

Virginia Polytechnic Institute and State University **Virginia, USA**
M.Sc. in Computer Science *GPA: 4.00/4.00 | January 2023 – May 2025*

Khulna University of Engineering & Technology **Khulna, Bangladesh**
B.Sc. in Computer Science and Engineering *CGPA: 3.77/4.00 | Rank: 7th/134 | March 2015 – March 2020*

PUBLICATIONS & PRESENTATIONS

1. **Tahmina Sultana Priya**, Huihuang Yan, Kirk J. Wangensteen, Stephen Wu, Anthony C. Luehrs, Filippo Pinto e Vairo, Fan Leng, Andres J. Acosta, Robert A. Vierkant, Alina M. Allen, Konstantinos N. Lazaridis, Eric W. Klee, Danfeng (Daphne) Yao, Shulan Tian ”Subtyping Metabolic Dysfunction-Associated Steatotic Liver Disease using Electronic Health Record-Linked Genomic Cohorts Reveals Diverse Etiologies and Progression” ***Accepted in Nature Communications, 2026***.
2. **Tahmina Sultana Priya**, Kirk J. Wangensteen, Eric W. Klee, Shulan Tian, Danfeng (Daphne) Yao ”Explainable and subgroup-specific machine learning models to predict rapid fibrosis progression in MASLD” ***Under Submission, 2026***.
3. Tanmoy Sarkar Pias, Wenjia Song, **Tahmina Sultana Priya**, Sahil Dudani, Randy Marchany, Brad Tilley, Lingxiang Wang, Danfeng (Daphne) Yao ”On the Readiness of Enterprise-scale Log Analysis Solutions Against Advanced Persistent Threats” ***Published in IEEE Secure Development (SecDev) Conference***, October 10th, 2025, Indianapolis, United States.
4. **Tahmina Sultana Priya**, Fan Leng, Anthony C. Luehrs, Eric W. Klee, Alina M. Allen, Konstantinos N. Lazaridis, Danfeng (Daphne)Yao, Shulan Tian ”Deep Phenotyping of Non-Alcoholic Fatty Liver Disease Patients with Genetic Factors for Insights into the Complex Disease.” ***Published in Machine Learning for Health (ML4H)***, December 10th, 2023, New Orleans, United States.
5. **Tahmina Sultana Priya**, Huihuang Yan, Stephen Wu, Filippo Pinto e Vairo, Anthony C. Luehrs, Fan Leng, Robert A. Vierkant, Eric W. Klee, Alina M. Allen, Konstantinos N. Lazaridis, Shulan Tian, Danfeng (Daphne) Yao ”Subtyping of metabolic dysfunction-associated steatotic liver disease (MASLD) by real-world integration of electric health records and population genomics data” - Presented at the 32nd Conference on **Intelligent Systems for Molecular Biology (ISMB)**, Montreal, QC, Canada; June 12-16, 2024; **Poster**
6. **Tahmina Sultana Priya**; Fan Leng; Anthony C. Luehrs; Huihuang Yan, Eric W. Klee; Alina M. Allen, Konstantinos N. Lazaridis, Shulan Tian, Danfeng (Daphne) Yao ”Latent class analysis-based identification of non-alcoholic fatty liver disease subtypes with distinct metabolic and genetic signatures” - Presented at the 20th Annual Meeting of **MCBIOS**, Emory University, Atlanta, GA, USA; March 22-24, 2024; **Lightning Talk**

RESEARCH EXPERIENCE

Virginia Polytechnic Institute and State University **Virginia, USA**
Graduate Research Assistant, Department of Computer Science *May 2023 - Present*

- Conduct healthcare data analytics using electronic health records (EHR), clinical, and genomic datasets to generate insights supporting disease characterization, patient outcomes, and precision healthcare research.
- Extract, clean, harmonize, and integrate large-scale healthcare datasets from EHR and genomic sources to identify clinically meaningful patient subgroups and characterize disease patterns.
- Develop and evaluate predictive and statistical models for disease prognosis, risk stratification, and healthcare outcome analysis in complex metabolic disorders.
- Develop explainable and subgroup-specific predictive models to identify clinical factors associated with disease progression and support interpretable healthcare decision-making.

Khulna University of Engineering & Technology **Khulna, Bangladesh**
Undergraduate Thesis (Supervised by: Dr. Muhammad Aminul Haque Akhand) *June 2019-Mar 2020*

- Addressed the NP-Hard problem of the Nurse Rostering Problem (NRP) with a focus on creating both valid and optimal solutions using a Modified Particle Swarm Optimization approach
- Achieved better results compared to various other techniques, leveraging the INRC-II dataset

WORK EXPERIENCE

Mayo Clinic **Rochester, MN, USA**
Research Intern – Bioinformatics, Division of Computational Biology
(Supervised by: Dr. Shulan Tian) *May 2025 – August 2025*
January 2024 – August 2024

- Analyzed electronic health records (EHR), clinical, and genomic datasets to extract meaningful healthcare insights and identify factors associated with disease progression.
- Performed data preprocessing, integration, validation, and statistical analysis of large-scale biomedical datasets.
- Developed predictive machine learning models for healthcare analytics, patient risk assessment, and outcome prediction.
- Identified clinically meaningful disease subgroups using multi-omics data integration and computational phenotyping approaches.
- Conducted survival analysis and drug response analysis to support precision medicine research and clinical decision-making.

Virginia Polytechnic Institute and State University **Virginia, USA**
Graduate Teaching Assistant, Department of Computer Science *January 2023 – December 2023*

- Helped students grasp core concepts in Data Structures and Algorithms, Computer Organization, and Android Development.
- Provided guidance on programming, system architecture, and mobile app development.
- Developed project solutions and graded assignments.

Daffodil International University **Dhaka, Bangladesh**
Lecturer, Department of Computer Science and Engineering *January 2021 – December 2022*

- Taught courses including Web Engineering, Data Mining, Machine Learning, Software Engineering, Digital Electronics, Software Project, Numerical Methods, and Data Communication.
- Instructed over 200 students and supervised 5 student project groups.

Vertical Innovations Limited **Dhaka, Bangladesh**
Senior Software Engineer *July 2022 – December 2022*
System Engineer *August 2020 – June 2022*

- Served as a team lead to build scalable web applications using Django and Nuxt.js.
- Developed data analytics dashboards and visualizations using Python.
- Integrated online payment gateways and deployed cloud-based infrastructure.
- Designed IoT-based data collection and monitoring systems and supported platform operations.

KEY SKILLS

Programming:	R, Python, C/C++, Java
Healthcare Analytics & ML:	Predictive Modeling, Statistical Learning, Survival Analysis, Risk Prediction, Explainable AI, PyTorch, TensorFlow, Keras, Scikit-learn
Healthcare Data:	Electronic Health Records (EHR), Genetic Data, Phenotyping
Databases:	SQL, Oracle, MySQL, PostgreSQL
Web Technologies:	HTML, CSS, PHP, JavaScript, Django, Vue.js, Nuxt.js, Laravel
Tools & Platforms:	Git, WEKA, Android Studio, VS Code, Eclipse, DigitalOcean

AWARDS & ACHIEVEMENTS

- Pratt Fellowship, Virginia Tech (2024–2025)
- Dean's Award, KUET (2017–2018, 2019–2020)
- 1st Runner-up, Textile Fest KUET – Fire-fighting Robot Project
- Government Scholarships, Bangladesh (General and Talent Pool)